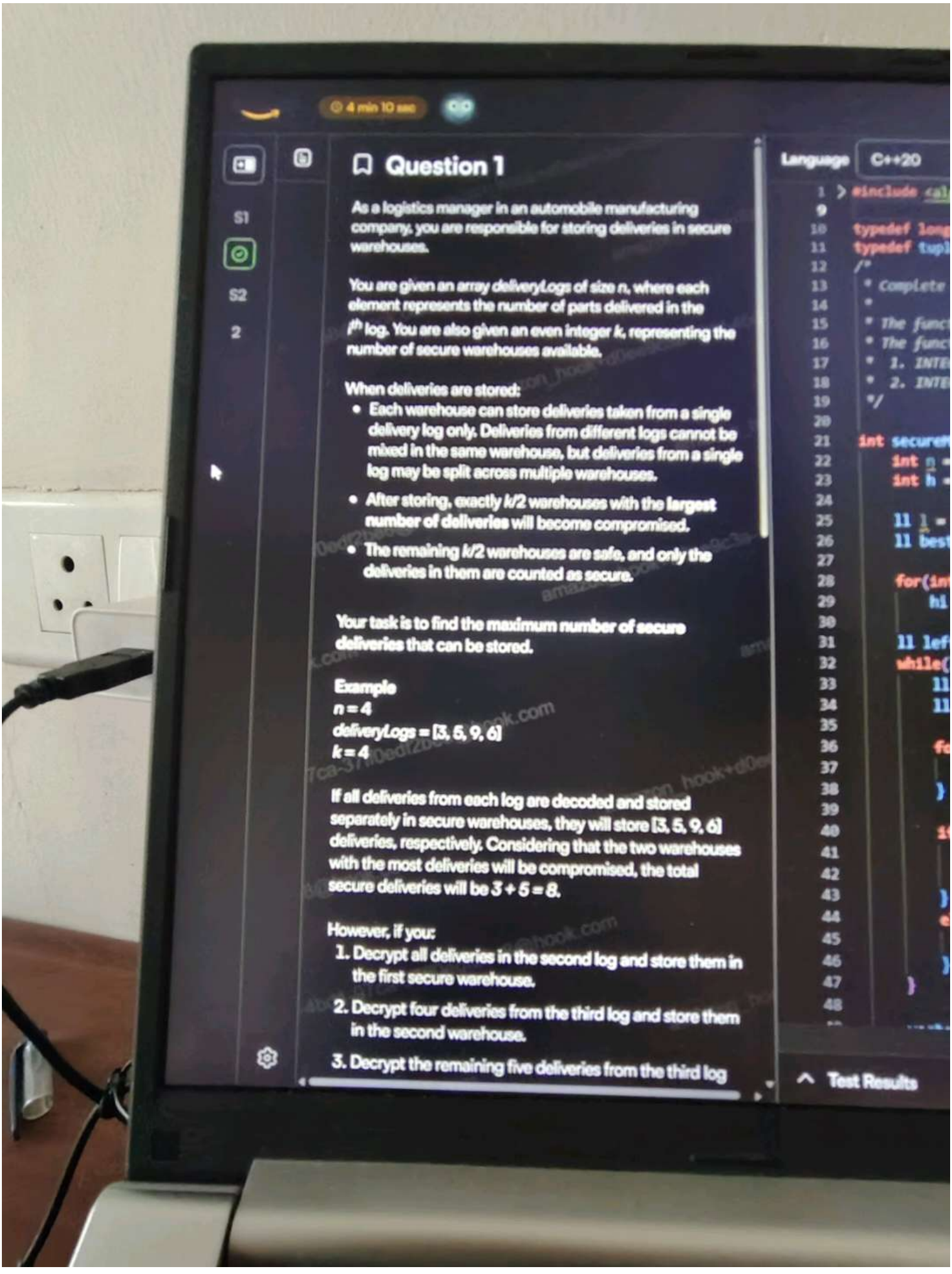
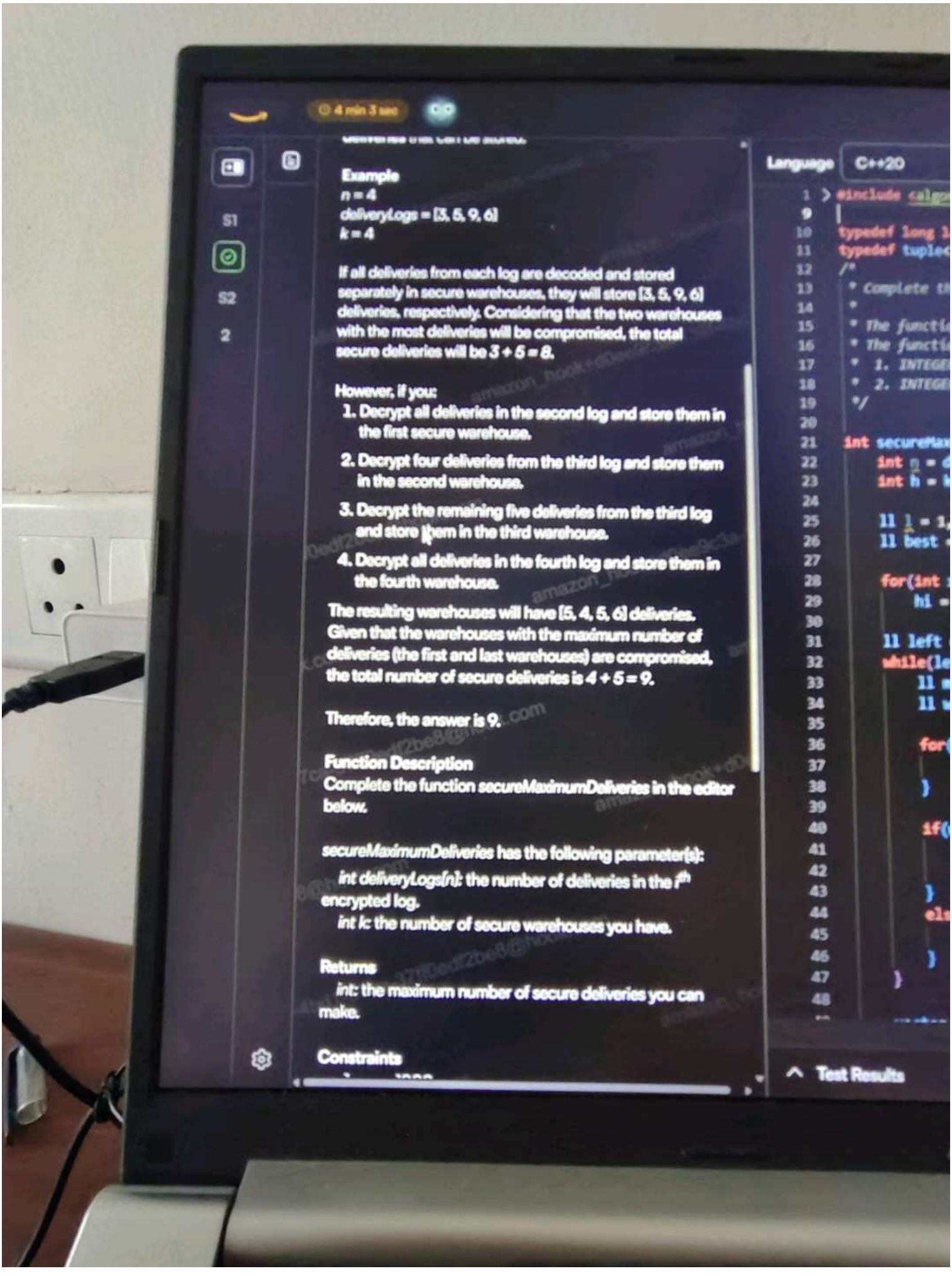
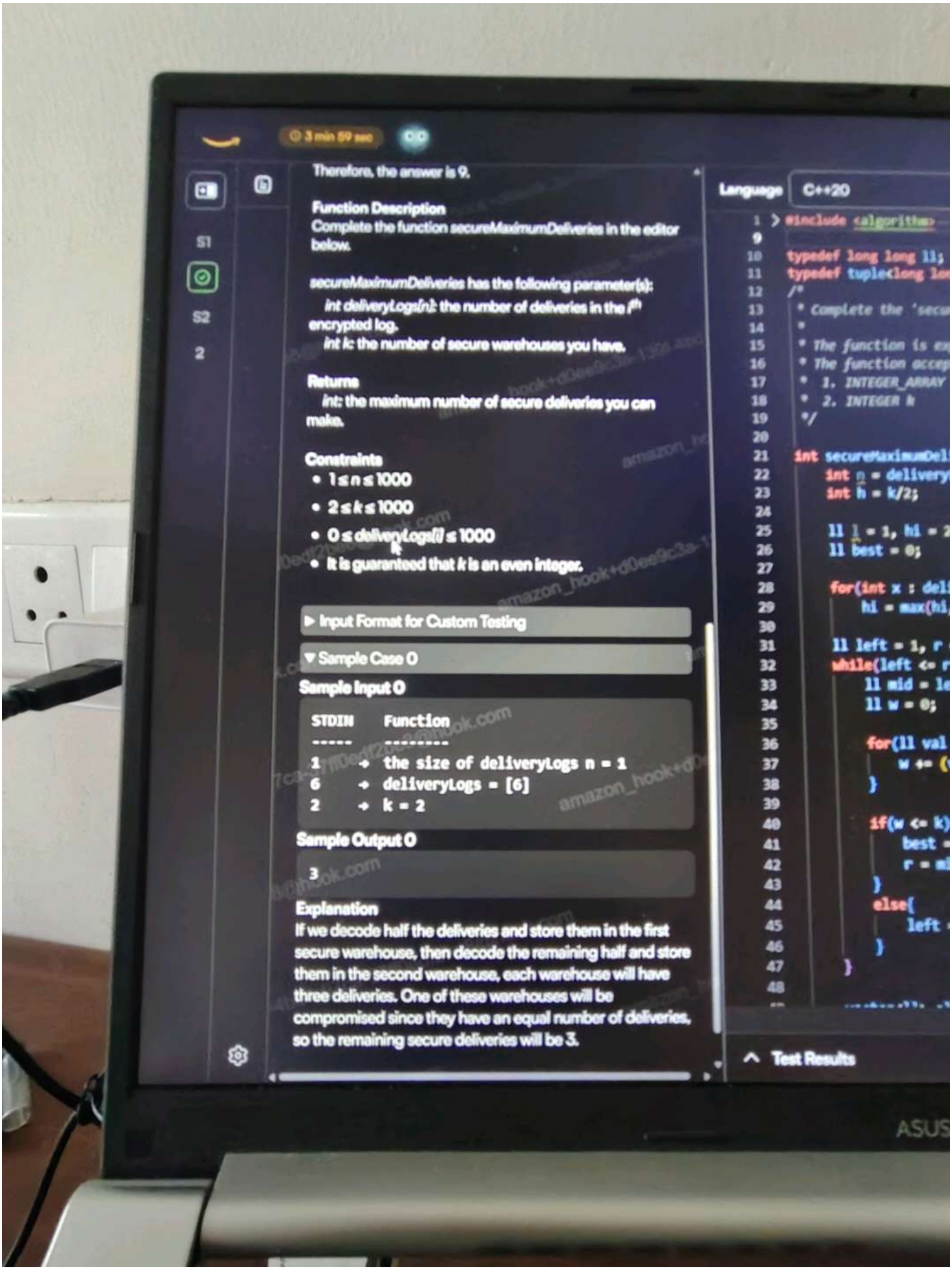
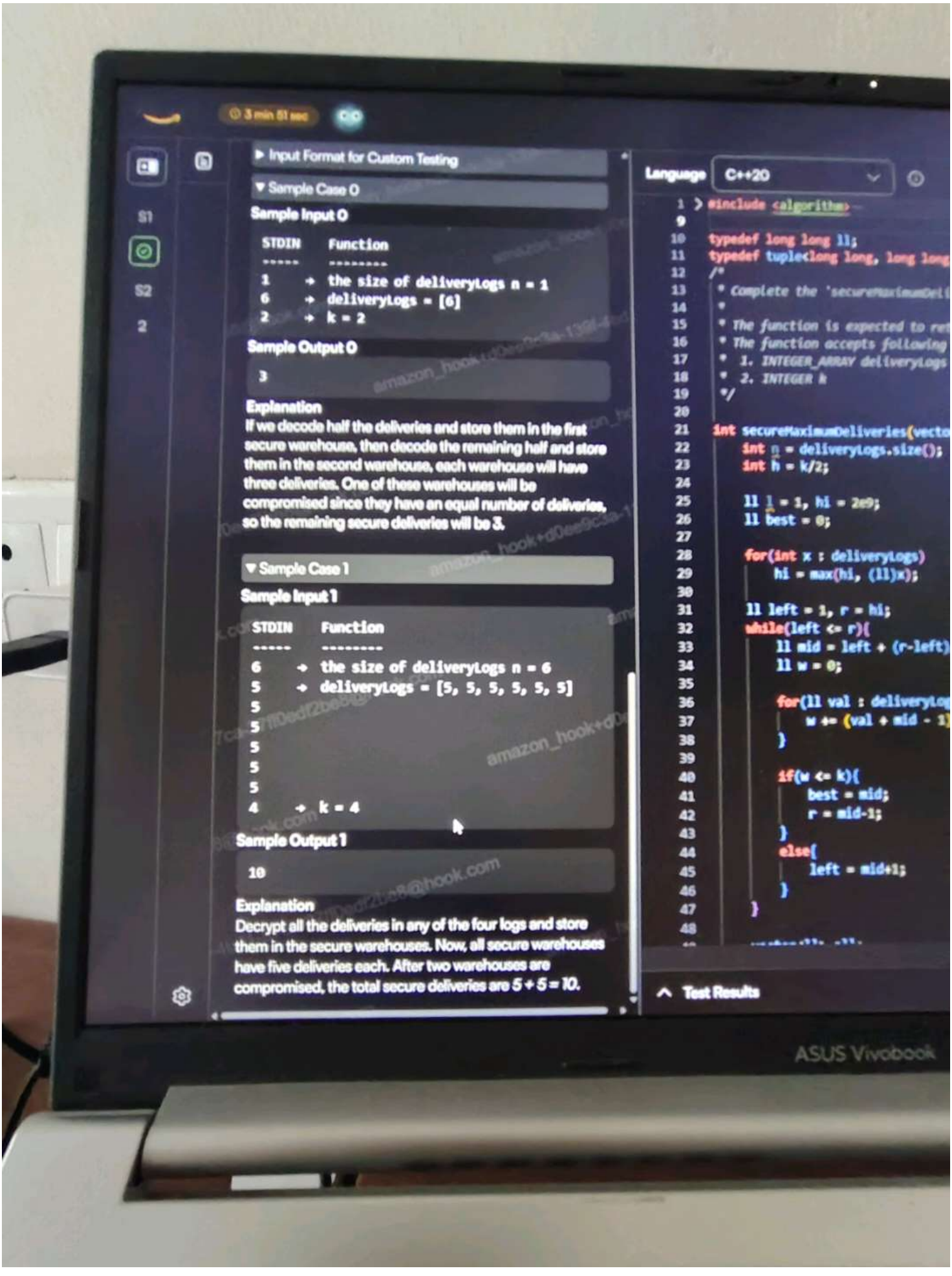

4 min 10 sec
S1
S2
2
Question 1
As a logistics manager in an automobile manufacturing company, you are responsible for storing deliveries in secure warehouses.
You are given an array deliveryLogs of size n, where each element represents the number of parts delivered in the ith log. You are also given an even integer k, representing the number of secure warehouses available.
When deliveries are stored:
Each warehouse can store deliveries taken from a single delivery log only. Deliveries from different logs cannot be mixed in the same warehouse, but deliveries from a single log may be split across multiple warehouses.
After storing, exactly k/2 warehouses with the largest number of deliveries will become compromised.
The remaining k/2 warehouses are safe, and only the deliveries in them are counted as secure.
Your task is to find the maximum number of secure deliveries that can be stored.
Example
n = 4
deliveryLogs = [3, 5, 9, 6]
k = 4
If all deliveries from each log are decoded and stored separately in secure warehouses, they will store [3, 5, 9, 6] deliveries, respectively. Considering that the two warehouses with the most deliveries will be compromised, the total secure deliveries will be 3 + 5 = 8.
However, if you:
1. Decrypt all deliveries in the second log and store them in the first secure warehouse.
2. Decrypt four deliveries from the third log and store them in the second warehouse.
3. Decrypt the remaining five deliveries from the third log
Language
C++20
Test Results


4 min 3 sec
S1
S2
2
Example
n = 4
deliveryLogs = [3, 5, 9, 6]
k = 4
If all deliveries from each log are decoded and stored separately in secure warehouses, they will store [3, 5, 9, 6] deliveries, respectively. Considering that the two warehouses with the most deliveries will be compromised, the total secure deliveries will be 3 + 5 = 8.
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1. Decrypt all deliveries in the second log and store them in the first secure warehouse.
2. Decrypt four deliveries from the third log and store them in the second warehouse.
3. Decrypt the remaining five deliveries from the third log and store them in the third warehouse.
4. Decrypt all deliveries in the fourth log and store them in the fourth warehouse.
The resulting warehouses will have [5, 4, 5, 6] deliveries. Given that the warehouses with the maximum number of deliveries (the first and last warehouses) are compromised, the total number of secure deliveries is 4 + 5 = 9.
Therefore, the answer is 9.
Function Description
Complete the function secureMaximumDeliveries in the editor below.
secureMaximumDeliveries has the following parameter(s):
int deliveryLogs[n]: the number of deliveries in the ith encrypted log.
int k: the number of secure warehouses you have.
Returns
int: the maximum number of secure deliveries you can make.
Constraints
Language
C++20
Test Results


3 min 59 sec
S1
S2
2
Therefore, the answer is 9.
Function Description
Complete the function secureMaximumDeliveries in the editor below.
secureMaximumDeliveries has the following parameter(s):
int deliveryLogs[n]: the number of deliveries in the ith encrypted log.
int k: the number of secure warehouses you have.
Returns
int: the maximum number of secure deliveries you can make.
Constraints
1 ≤ n ≤ 1000
2 ≤ k ≤ 1000
0 ≤ deliveryLogs[i] ≤ 1000
It is guaranteed that k is an even integer.
Input Format for Custom Testing
Sample Case 0
Sample Input 0
STDIN Function
1 → the size of deliveryLogs n = 1
6 → deliveryLogs = [6]
2 → k = 2
Sample Output 0
3
Explanation
If we decode half the deliveries and store them in the first secure warehouse, then decode the remaining half and store them in the second warehouse, each warehouse will have three deliveries. One of these warehouses will be compromised since they have an equal number of deliveries, so the remaining secure deliveries will be 3.
Language
C++20
Test Results


3 min 51 sec
Input Format for Custom Testing
Sample Case 0
Sample Input 0
STDIN Function
1 → the size of deliveryLogs n = 1
6 → deliveryLogs = [6]
2 → k = 2
Sample Output 0
3
Explanation
If we decode half the deliveries and store them in the first secure warehouse, then decode the remaining half and store them in the second warehouse, each warehouse will have three deliveries. One of these warehouses will be compromised since they have an equal number of deliveries, so the remaining secure deliveries will be 3.
Sample Case 1
Sample Input 1
STDIN Function
6 → the size of deliveryLogs n = 6
5 → deliveryLogs = [5, 5, 5, 5, 5, 5]
4 → k = 4
Sample Output 1
10
Explanation
Decrypt all the deliveries in any of the four logs and store them in the secure warehouses. Now, all secure warehouses have five deliveries each. After two warehouses are compromised, the total secure deliveries are 5 + 5 = 10.
Language C++20
#include <algorithm>
typedef long long ll;
typedef tuple<long long, long long
/*
* Complete the 'secureMaximumDeli
*
* The function is expected to re
* The function accepts following
* 1. INTEGER_ARRAY deliveryLogs
* 2. INTEGER k
*/
int secureMaximumDeliveries(vecto
int n = deliveryLogs.size();
int h = k/2;
ll l = 1, hi = 2e9;
ll best = 0;
for(int x : deliveryLogs)
hi = max(hi, (ll)x);
ll left = 1, r = hi;
while(left <= r){
ll mid = left + (r-left)
ll w = 0;
for(ll val : deliveryLo
w += (val + mid - 1
}
if(w <= k){
best = mid;
r = mid-1;
}
else{
left = mid+1;
}
}
Test Results
ASUS Vivobook